

SOC Pipeline Implementation using Wazuh, Logstash and Elasticsearch

This document explains the implementation of a security monitoring pipeline using Wazuh inside a VirtualBox lab environment. The lab includes a Wazuh server, a Kali Linux endpoint with a Wazuh agent, and a Logstash pipeline forwarding alerts into Elasticsearch.

1. Lab Environment

- 1 Oracle VirtualBox used as virtualization platform
- 2 Wazuh server installed using downloadable installer
- 3 Kali Linux VM used as monitored endpoint
- 4 Wazuh agent installed on Kali Linux
- 5 Logstash installed to process alerts
- 6 Elasticsearch / Wazuh Indexer used for log storage

2. Architecture

Kali Linux (Wazuh Agent)

↓

Wazuh Manager

↓ alerts.json

↓

Logstash Pipeline

↓

Elasticsearch

↓

Dashboard Visualization

3. Virtual Machine Setup

- 1 Install Oracle VirtualBox
- 2 Create Wazuh server virtual machine
- 3 Create Kali Linux virtual machine

- 4 Ensure both machines are on the same network

4. Wazuh Installation

The Wazuh server was installed using the downloadable installation package. The installation deploys Wazuh manager, indexer, and dashboard components.

- 1 Wazuh Manager
- 2 Wazuh Indexer (OpenSearch)
- 3 Wazuh Dashboard
- 4 Filebeat log shipper

5. Accessing Dashboard

```
https:// ip add
```

6. Installing Wazuh Agent on Kali

```
sudo apt update  
sudo apt  
install wazuh-agent
```

```
sudo nano /var/ossec/etc/ossec.conf
```

```
WAZUH_SERVER_IP
```

```
sudo systemctl enable wazuh-agent  
sudo systemctl start wazuh-agent
```

7. Wazuh Alerts Location

```
/var/ossec/logs/alerts/alerts.json
```

8. Installing Logstash

```
sudo apt update
```

```
sudo apt install logstash
```

```
sudo systemctl enable logstash sudo systemctl start logstash
```

9. Logstash Pipeline Configuration

```
sudo nano /etc/logstash/conf.d/wazuh.conf
```

```
input { file { path =>
"/var/ossec/logs/alerts/alerts.json"
start_position => "beginning"
sourcedb_path => "/dev/null" codec
=> json } }
```

```
filter { mutate { add_field => {
"log_source" => "wazuh" } } }
```

```
output { stdout {
codec => rubydebug
}
```

```
elasticsearch { hosts =>
["https://localhost:9200"]
user => "admin" password =>
"admin" ssl => true
ssl_certificate_verification
=> false index =>
"wazuhagent-logs" }
}
```

10. Testing Configuration

```
sudo /usr/share/logstash/bin/logstash --path.settings /etc/logstash -f  
/etc/logstash/conf.d/wazuh.conf --config.test_and_exit
```

11. Running Logstash

```
sudo /usr/share/logstash/bin/logstash --path.settings /etc/logstash -f  
/etc/logstash/conf.d/wazuh.conf
```

12. Issues Encountered

- 1 Terminal paste not working inside virtual machine
- 2 Incorrect command formatting
- 3 Logstash pipeline syntax errors
- 4 Pipeline shutdown during testing

13. Conclusion

This lab demonstrated how Wazuh agents collect logs from endpoints and how Logstash can forward those alerts into Elasticsearch. The project provides hands-on experience with Security Operations Center monitoring architecture.

```

Version 2023.10.20260216:
Run the following command to upgrade to 2023.10.20260216:

dnf upgrade --releasever=2023.10.20260216

Release notes:
https://docs.aws.amazon.com/linux/a12023/release-notes/relnotes-2023.10.20260216.html

Version 2023.10.20260302:
Run the following command to upgrade to 2023.10.20260302:

dnf upgrade --releasever=2023.10.20260302

Release notes:
https://docs.aws.amazon.com/linux/a12023/release-notes/relnotes-2023.10.20260302.html
=====

Installed:
logstash-1:8.19.12-1.x86_64

Complete!
[wazuh-user@wazuh-server ~]$ sudo systemctl start logstash
[wazuh-user@wazuh-server ~]$ sudo systemctl status logstash
■ logstash.service - logstash
   Loaded: loaded (/usr/lib/systemd/system/logstash.service; disabled; preset: disabled)
   Active: active (running) since Wed 2026-03-11 09:33:28 UTC; 7s ago
     Main PID: 5423 (java)
       Tasks: 24 (limit: 9478)
      Memory: 354.0M
         CPU: 21.602s
    CGroup: /system.slice/logstash.service
            └─5423 /usr/share/logstash/jdk/bin/java -Xms1g -Xmx1g -Djava.awt.headless=true -Dfile.

Mar 11 09:33:28 wazuh-server logstash[5423]: Using bundled JDK: /usr/share/logstash/jdk
[wazuh-user@wazuh-server ~]$

```

```

NAME="Amazon Linux"
VERSION="2023"
ID="amzn"
ID_LIKE="fedora"
VERSION_ID="2023"
PLATFORM_ID="platform:a12023"
PRETTY_NAME="Amazon Linux 2023.10.20260120"
ANSI_COLOR="0;33"
CPE_NAME="cpe:2.3:o:amazon:amazon_linux:2023"
HOME_URL="https://aws.amazon.com/linux/amazon-linux-2023/"
DOCUMENTATION_URL="https://docs.aws.amazon.com/linux/"
SUPPORT_URL="https://aws.amazon.com/premiumsupport/"
BUG_REPORT_URL="https://github.com/amazonlinux/amazon-linux-2023"
VENDOR_NAME="AWS"
VENDOR_URL="https://aws.amazon.com/"
SUPPORT_END="2029-06-30"
[wazuh-user@wazuh-server ~]$ sudo rpm --import
rpmkeys: no arguments given
[wazuh-user@wazuh-server ~]$ sudo rpm --import https://artifacts.elastic.co/GPG-KEY-elasticsearch
[wazuh-user@wazuh-server ~]$ sudo tee /etc/yum.repos.d/elastic.repo <<EOF
> [elastic-8.x]
> name= Elastic repository for 8.x packages
> baseurl=https://artifacts.elastic.co/packages/8.x/yum
> gpgkey=https://artifacts.elastic.co/GPG-KEY-elasticsearch
> enabled=1
> autorefresh=1
> type=rpm-md
> EOF
[elastic-8.x]
name= Elastic repository for 8.x packages
baseurl=https://artifacts.elastic.co/packages/8.x/yum
gpgkey=https://artifacts.elastic.co/GPG-KEY-elasticsearch
enabled=1
autorefresh=1
type=rpm-md
[wazuh-user@wazuh-server ~]$ sudo dnf install logstash -y
Elastic repository for 8.x pac 57% [=====] 1 8.3 MB/s | 86 MB 00:07 ETA

```

```

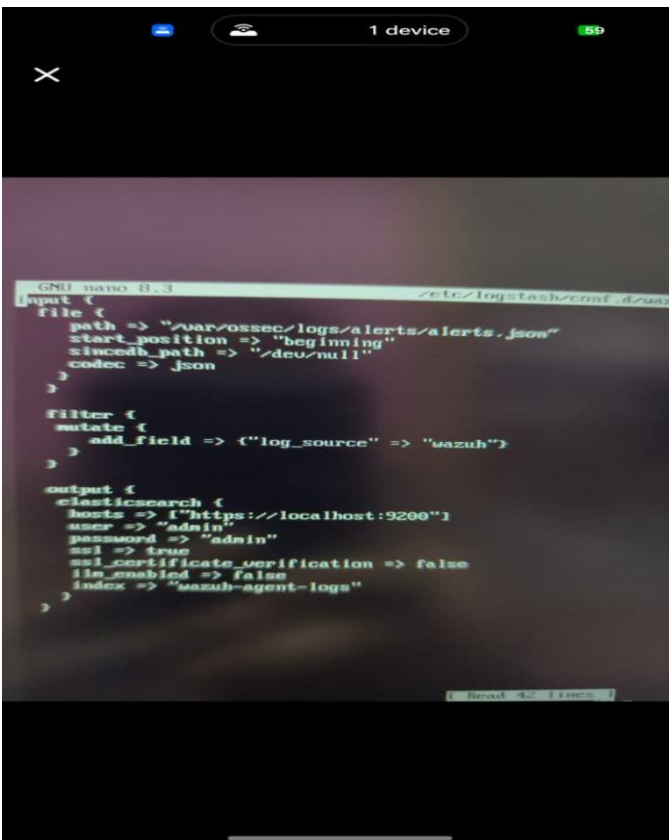
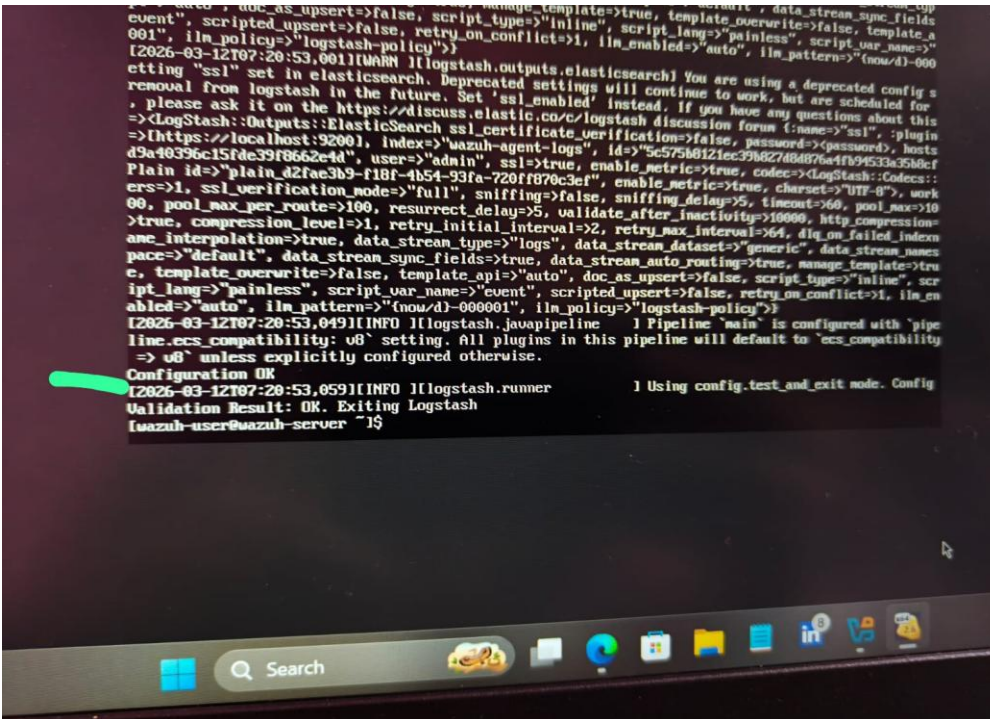
Tasks: 94 (limit: 9478)
Memory: 4.6G
CPU: 1min 45.399s
CGroup: /system.slice/wazuh-indexer.service
└─2097 /usr/share/wazuh-indexer/jdk/bin/java -Xshare:auto -Dopensearch.networkaddress.
Mar 11 12:22:07 wazuh-server systemd-entrypoint[2097]: WARNING: A terminally deprecated method in j
Mar 11 12:22:07 wazuh-server systemd-entrypoint[2097]: WARNING: System::setSecurityManager has been
Mar 11 12:22:07 wazuh-server systemd-entrypoint[2097]: WARNING: Please consider reporting this to t
Mar 11 12:22:07 wazuh-server systemd-entrypoint[2097]: WARNING: System::setSecurityManager will be
Mar 11 12:22:11 wazuh-server systemd-entrypoint[2097]: Mar 11, 2026 12:22:11 PM sun.util.locale.pro
Mar 11 12:22:11 wazuh-server systemd-entrypoint[2097]: WARNING: COMPAT locale provider will be remo
Mar 11 12:22:14 wazuh-server systemd-entrypoint[2097]: WARNING: A terminally deprecated method in j
Mar 11 12:22:14 wazuh-server systemd-entrypoint[2097]: WARNING: System::setSecurityManager has been
Mar 11 12:22:14 wazuh-server systemd-entrypoint[2097]: WARNING: Please consider reporting this to t
Mar 11 12:22:14 wazuh-server systemd-entrypoint[2097]: WARNING: System::setSecurityManager will be

[wazuh-user@wazuh-server cdrom]$ curl http://localhost:9200 -k
curl: (52) Empty reply from server
[wazuh-user@wazuh-server cdrom]$ curl -k -u admin:admin https://localhost:9200
{
  "name" : "node-1",
  "cluster_name" : "wazuh-cluster",
  "cluster_uuid" : "GFpQR5KdSXkxUPSFUryQ3g",
  "version" : {
    "number" : "7.10.2",
    "build_type" : "rpm",
    "build_hash" : "e27d4927fd308f52329c0644b83ca84987a68043",
    "build_date" : "2026-02-06T10:55:43.749460719Z",
    "build_snapshot" : false,
    "lucene_version" : "9.12.3",
    "minimum_wire_compatibility_version" : "7.10.0",
    "minimum_index_compatibility_version" : "7.0.0"
  },
  "tagline" : "The OpenSearch Project: https://opensearch.org/"
}
[wazuh-user@wazuh-server cdrom]$

```

green	open	wazuh-statistics-2026.11w	0	423.7kb	423.7kb	uuX4jloXS1SJeH53jIqIFQ	1	0	
green	open	wazuh-states-inventory-protocols-wazuh-server	3	0	8kb	8kb	nunpiWBJSo2KC4wA8EKnxA	1	0
green	open	wazuh-alerts-4.x-2026.03.09	67	0	451.1kb	451.1kb	WTFQGS0PTB2Mb64IAmkFuy	3	0
green	open	wazuh-alerts-4.x-2026.03.06	260	0	992.9kb	992.9kb	PyqidLLlQgih9If7Xxyiou	3	0
green	open	wazuh-monitoring-2026.11w	49	0	105.1kb	105.1kb	biJ77aHCTSyHsrB0C6aqGA	1	0
green	open	wazuh-alerts-4.x-2026.03.07	78	0	403.7kb	403.7kb	I8mIF2KQRBnEVfTFtILh8w	3	0
green	open	.kibana_1	9	2	33.8kb	33.8kb	Yt0oY64LTNuP547KvlfxoA	1	0
green	open	wazuh-states-inventory-services-wazuh-server	132	14	152.7kb	152.7kb	6SATn5MBR8ubDWJPKyHduw	1	0
green	open	wazuh-alerts-4.x-2026.03.11	218	0	1.1mb	1.1mb	Ga4dQk_jSDqkU3sUsRCyig	3	0
green	open	.plugins-m1-config	1	0	4kb	4kb	RRpwIWtNq5CQJ2M2nEhMdg	1	0
green	open	wazuh-alerts-4.x-2026.03.10	339	0	1mb	1mb	nB9CjWQ4RPC-YsI2EaxDCQ	3	0
green	open	wazuh-states-inventory-hardware-wazuh-server	1	0	8.1kb	8.1kb	EuW3q3hLRceSX2_X7aAvdQ	1	0
green	open	wazuh-states-inventory-interfaces-wazuh-server	2	0	10.4kb	10.4kb	X_dle5X3TASeSzFJewUpEA	1	0
green	open	wazuh-states-vulnerabilities-wazuh-server	1	0	21.5kb	21.5kb	OoLlVAYPT70SA1IS1bsLuQ	1	0
green	open	wazuh-states-inventory-networks-wazuh-server	3	0	12kb	12kb	l-EuugR2T92beWhbdwrc8A	1	0
green	open	wazuh-statistics-2026.10w	54	0	104.1kb	104.1kb	rDN8px24RGeRSLmb9FkzUw	1	0
green	open	wazuh-states-inventory-groups-wazuh-server	95	0	32.7kb	32.7kb	40giXueBQHS0xPfCpz79PQ	1	0
green	open	wazuh-states-inventory-processes-wazuh-server	279	0	58kb	58kb	lMuAMPGXt4ePSbuMcQ6AXA	1	0

[wazuh-user@wazuh-server cdrom]\$ █





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```
green open wazuh-states-vulnerabilities-wazuh-server 0 21.5kb 21.5kb 0oL1UAYFT7DSW11S1hcLuQ 1 0
green open wazuh-states-inventory-networks-wazuh-server 0 12kb 12kb 1-2mgKGT32he0hhdurc0h 1 0
green open wazuh-statistics-2025.10w 0 104.1kb 104.1kb rD80px24RGeRILab5Fkz0u 1 0
green open wazuh-states-inventory-groups-wazuh-server 0 32.7kb 32.7kb 40giXueBQ6S0xPFCpc75PQ 1 0
green open wazuh-states-inventory-processes-wazuh-server 0 50kb 50kb 1RwVFGK74cP3bAcQ60Xh 1 0
root@wazuh-server ~# sudo -u logstash cat /var/ossec/logs/alerts/alerts.json | head
cat: /var/ossec/logs/alerts/alerts.json: Permission denied
root@wazuh-server ~# (15233.562768) psouse serial: Explorer Mouse at isa0066/serial/input0 lost s
ynchronization, throwing 2 bytes away.
root@wazuh-server ~# (15240.793709) 16:35:33.024494 control Session 0 is about to close ...
(15240.0309429) 16:35:33.062315 control Stopping all guest processes ...
(15240.0327481) 16:35:33.067002 control Closing all guest files ...
(15241.9146481) e1000: eth0 NIC Link is Down
(15245.0400911) 05:49:37.000107 timesync vgsucTimeSyncMarker: Radical host time change: 47 664 790 00
0 000ns (GuestNow=1 773 294 577 870 000 000 ns HostLast=1 773 246 913 138 000 000 ns)
(15246.0043521) e1000: eth0 NIC Link is Up 1000 Mbps Full Duplex, Flow Control: RX
(15255.0401541) 05:49:47.004304 timesync vgsucTimeSyncMarker: Radical guest time change: 47 649 811 3
00 000ns (GuestNow=1 773 294 507 004 350 000 ns GuestLast=1 773 246 930 072 970 000 ns (SetTimeLastL
ong=true)
root@wazuh-server ~#
root@wazuh-server ~# sudo -u logstash cat /var/ossec/logs/alerts/alerts.json | head
cat: /var/ossec/logs/alerts/alerts.json: Permission denied
root@wazuh-server ~# sudo chmod 755 /var/ossec/logs/
root@wazuh-server ~# sudo chmod 755 /var/ossec/logs/alerts/
root@wazuh-server ~# sudo chmod 644 /var/ossec/logs/alerts/alerts.json
root@wazuh-server ~# sudo -u logstash cat /var/ossec/logs/alerts/alerts.json | head
cat: /var/ossec/logs/alerts/alerts.json: Permission denied
root@wazuh-server ~# ^C
root@wazuh-server ~#
```



Search





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```
commons-compiler-3.1.0.jar:/usr/share/logstash/logstash-core/lib/jars/commons-logging-1.2.5.jar:/usr/share/logstash/logstash-core/lib/jars/error_prone_annotations-2.26.1.jar:/usr/share/logstash/logstash-core/lib/jars/failureaccess-1.0.2.jar:/usr/share/logstash/logstash-core/lib/jars/guava-33.1.0-jre.jar:/usr/share/logstash/logstash-core/lib/jars/httpclient-4.5.14.jar:/usr/share/logstash/logstash-core/lib/jars/httpcore-4.4.14.jar:/usr/share/logstash/logstash-core/lib/jars/jackson-annotations-2.16.2.jar:/usr/share/logstash/logstash-core/lib/jars/jackson-core-2.16.2.jar:/usr/share/logstash/logstash-core/lib/jars/jackson-databind-2.16.2.jar:/usr/share/logstash/logstash-core/lib/jars/jackson-dataformat-cbor-2.16.2.jar:/usr/share/logstash/logstash-core/lib/jars/janino-3.1.0.jar:/usr/share/logstash/logstash-core/lib/jars/jar305-3.0.2.jar:/usr/share/logstash/logstash-core/lib/jars/jvm-options-parser-0.19.12.jar:/usr/share/logstash/logstash-core/lib/jars/enablefuture-9999.0-empty-to-avoid-conflict-with-guava.jar:/usr/share/logstash/logstash-core/lib/jars/log4j-1.2-api-2.17.2.jar:/usr/share/logstash/logstash-core/lib/jars/log4j-core-2.17.2.jar:/usr/share/logstash/logstash-core/lib/jars/log4j-jcl-2.17.2.jar:/usr/share/logstash/logstash-core/lib/jars/log4j-slf4j-impl-2.17.2.jar:/usr/share/logstash/logstash-core/lib/jars/logstash-core.jar:/usr/share/logstash/logstash-core/lib/jars/slf4j-api-1.7.32.jar:/usr/share/logstash/logstash-core/lib/jars/snakeyaml-2.2.jar org.logstash.Logstash --path.settings /etc/logstash.yml
root 42391 0.0 0.0 222328 2100 pts/3 R+ 16:27 0:00 grep --color=auto logst
froot@wazuh-server ~# curl -k -u admin:admin https://localhost:9200
{
  "name" : "node-1",
  "cluster_name" : "wazuh-cluster",
  "cluster_uuid" : "GfPQf5K4S0xVPSF0yQ3y",
  "version" : {
    "number" : "7.10.2",
    "build_type" : "rpm",
    "build_hash" : "e27d4927fd308f52329c0644b83ca84987a68043",
    "build_date" : "2026-02-06T10:55:43.749460719Z",
    "build_snapshot" : false,
    "lucene_version" : "9.12.3",
    "minimum_wire_compatibility_version" : "7.10.0",
    "minimum_index_compatibility_version" : "7.0.0"
  },
  "tagline" : "The OpenSearch Project: https://opensearch.org/"
}
```



Search

